CSCI3340 Scientific Computing

2025–26 Term 2

Assignment 1

Submission Deadline: 23:59:00, 5th February 2026.

Submission Guidelines

Assignment 1 covers material from Lectures 1–3 and is worth a total of 100 points. There are 6 problems in total: the first 3 are written questions, and the remaining 3 are programming tasks.

You should submit a single `.zip` file containing exactly one PDF file and all relevant MATLAB files. The PDF must include your solutions to all problems and should be self-contained, clear, and easy to read. You may write your answers by hand or type them. For each programming problem, clearly indicate in your PDF which MATLAB file corresponds to that problem. Your PDF should also include brief instructions on how to run your code and interpret the results.

If the submission was late but within 3 days after the deadline, a -30 points penalty will be applied. Submissions made after this period will not be graded.

If you have any concerns, feel free to contact me (colintsang@cuhk.edu.hk) or the TA in charge (srijan25@link.cuhk.edu.hk).

Problem 1 (10 points)

(a) From the *standard model of arithmetic*, we know that for $a, b \in \mathbb{M}$, we have

$$a \tilde{-} b = (a - b)(1 + r)$$

where r is the relative rounding error with $|r| < \epsilon$. We denote with $-$ the exact subtraction and with $\tilde{-}$ the equivalent machine subtraction. Prove that the subtraction operation satisfies Wilkinson's Principle.

(b) Consider the problem of subtraction $\mathcal{P}(x_1, x_2) := x_1 - x_2$. Determine the condition number if

(i) $x_1 x_2 < 0$, i.e., the two values have opposite signs.

(ii) $x_1 x_2 > 0$ and $x_1 \approx x_2$.

✓ **Problem 2 (10 points)**

Consider the matrix

$$A = \begin{pmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{pmatrix}.$$

Since A is symmetric positive definite, there exists an upper triangular matrix R such that $A = R^T R$. In our lecture, we showed that the j-th row of R, $r_{j:}$, are given by $r_{j:} = \frac{v^T}{\sqrt{v_j}}$, where

$$v^T := a_{j:} - \sum_{k=1}^{j-1} r_{kj} r_{k:}$$

Use the above formula to compute each row of R. Write down all the computation steps.

Problem 3 (20 points)

Consider the equation

$$\frac{\cos(x)}{x} = 1$$

which has a solution in the interval $[0, 1]$.

(a) Rewrite this equation in the form $x = g(x)$ suitable for fixed-point iteration.

(b) Show that $g(x)$ maps the interval $[0, 1]$ into $[0, 1]$.

(c) Starting from $x_0 = 0.5$, perform three iterations of the fixed-point method $x_{n+1} = g(x_n)$. Write down each step.

(d) Discuss briefly whether the fixed-point iteration is expected to converge for this function and interval. Justify your answer by examining $|g'(x)|$ on $[0, 1]$.

✓ **Problem 4 (20 points)**

The function

$$f(x) = \frac{x^2}{1!} + 7\frac{x^4}{2!} + 17\frac{x^6}{3!} + \cdots = \sum_{n=1}^{\infty} (2n^2 - 1)\frac{x^{2n}}{n!}$$

should be evaluated to machine precision for $x \in (0, 25)$. Write a MATLAB function for this purpose. Your code should compute the result to machine precision with an elegant stopping criterion. It should also avoid any potential overflows.

Hints: Consider the n-th term $T_n = (2n^2 - 1)\frac{x^{2n}}{n!}$. You should compute T_n recursively to avoid direct computation of x^{2n} and $n!$, which may overflow for large n. You may try to obtain a recurrence relation by simplifying $\frac{T_n}{T_{n-1}}$.

✓ **Problem 5 (20 points)**

Write a MATLAB function from scratch that solves a linear system $A\mathbf{x} = \mathbf{b}$ using LU factorization with partial pivoting. Your function should accept a square matrix A and a vector $\mathbf{b}$ as inputs, and then return the solution vector $\mathbf{x}$.

✓ **Problem 6 (20 points)**

An open-top box is to be made by cutting squares of side length x from each corner of a 10×16 inch rectangular sheet of metal and folding up the sides. Determine the value(s) of x such that the volume of the box is exactly 100 cubic inches. Solve by implementing Newton's method in MATLAB.

Problem 1 (10 points)

(a) From the *standard model of arithmetic*, we know that for $a, b \in \mathbb{M}$, we have

$$a \tilde{-} b = (a - b)(1 + r)$$

where r is the relative rounding error with $|r| < \epsilon$. We denote with $-$ the exact subtraction and with $\tilde{-}$ the equivalent machine subtraction. Prove that the subtraction operation satisfies Wilkinson's Principle.

proof:

$a \tilde{-} b = (a-b)(1+r) = (1+r)a - (1+r)b = \tilde{a} - \tilde{b}$

where $\tilde{a} = (1+r)a$, $\tilde{b} = (1+r)b$, which are slightly pretubed result of a, b

that is, the result of numerical subtraction on computer is the result of an exact computation with slightly perturbed initial data.

which proves the Wilkinson's Principle.

(b) Consider the problem of subtraction $\mathcal{P}(x_1, x_2) := x_1 - x_2$. Determine the condition number if

$|a+b| \le |a| + |b|$

(i) $x_1 x_2 < 0$, i.e., the two values have opposite signs.

(ii) $x_1 x_2 > 0$ and $x_1 \approx x_2$.

Let the perturbation of x_1 and x_2 be $\hat{x}_1 = x_1(1+\varepsilon_1)$; $\hat{x}_2 = x_2(1+\varepsilon_2)$ where $|\varepsilon_1|, |\varepsilon_2| < \varepsilon$

$$\Rightarrow \frac{\|\hat{P}(x_1, x_2) - P(x_1, x_2)\|}{\|P(x_1, x_2)\|} = \left| \frac{[x_1(1+\varepsilon_1) - x_2(1+\varepsilon_2)] - (x_1 - x_2)}{x_1 - x_2} \right| = \left| \frac{\varepsilon_1 x_1 - \varepsilon_2 x_2}{x_1 - x_2} \right|$$

$$= \frac{\varepsilon_1(x_1 - x_2) + \varepsilon_1 x_2 - \varepsilon_2 x_2}{x_1 - x_2} = \varepsilon_1 + \frac{x_2}{x_1 - x_2}(\varepsilon_1 - \varepsilon_2)$$

(i) when $x_1 x_2 < 0 \Rightarrow |x_1 - x_2| = |x_1| + |x_2|$

$$\Rightarrow \left| \frac{\varepsilon_1 x_1 - \varepsilon_2 x_2}{x_1 - x_2} \right| = \left| \frac{x_1}{x_1 - x_2}\varepsilon_1 - \frac{x_2}{x_1 - x_2}\varepsilon_2 \right| = \left|\frac{x_1}{x_1 - x_2}\right| \varepsilon_1 + \left|\frac{x_2}{x_1 - x_2}\right| \varepsilon_2 = \frac{|x_1|}{|x_1| + |x_2|}\varepsilon_1 + \frac{|x_2|}{|x_1| + |x_2|}\varepsilon_2$$

$$\approx \frac{|x_1| + |x_2|}{|x_1| + |x_2|}\varepsilon = \varepsilon$$

$\Rightarrow K = 1$

(ii) Since $x_1 \approx x_2$, the condition number could reach infinity ∞, which is the catestrophic cancellation.

Problem 2 (10 points)

Consider the matrix

$$A = \begin{pmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{pmatrix}.$$

Since A is symmetric positive definite, there exists an upper triangular matrix R such that $A = R^T R$. In our lecture, we showed that the j-th row of R, $r_{j:}$, are given by $r_{j:} = \frac{v^T}{\sqrt{v_j}}$, where

$$v^T := a_{j:} - \sum_{k=1}^{j-1} r_{kj} r_{k:}$$

Use the above formula to compute each row of R. Write down all the computation steps.

① when $j=1$,

$v^T = a_{1:} - \sum_{k=1}^{0} r_{k1} r_{k:} = a_{1:} = (4 \quad 12 \quad -16)$

$v_1 = 4$

$\Rightarrow r_{1:} = \frac{v^T}{\sqrt{v_1}} = \frac{(4 \quad 12 \quad -16)}{2} = (2 \quad 6 \quad -8)$

②. when $j=2$,

$v^T = a_{2:} - \sum_{k=1}^{1} r_{k2} r_{k:} = a_{2:} - r_{12} r_{1:} = (12 \quad 37 \quad -43) - \underline{6 * (2 \quad 6 \quad -8)} = (0 \quad 1 \quad 5)$

$r_{2:} = \frac{v^T}{\sqrt{v_2}} = \frac{(0 \quad 1 \quad 5)}{1} = (0 \quad 1 \quad 5)$

③. when $j=3$

$v^T = a_{3:} - \sum_{k=1}^{2} r_{k3} r_{k:} = a_{3:} - r_{13} r_{1:} - r_{23} r_{2:} = (-16 \quad -43 \quad 98) - (-8) * (2 \quad 6 \quad -8) - 5 * (0 \quad 1 \quad 5)$

$= (0 \quad 0 \quad 9)$

$r_{3:} = \frac{v^T}{\sqrt{v_3}} = \frac{(0 \quad 0 \quad 9)}{3} = (0 \quad 0 \quad 3)$

$$\Rightarrow R = \begin{pmatrix} 2 & 6 & -8 \\ 0 & 1 & 5 \\ 0 & 0 & 3 \end{pmatrix}$$

Problem 3 (20 points)

Consider the equation

$$\frac{\cos(x)}{x} = 1$$

which has a solution in the interval $[0, 1]$.

(a) Rewrite this equation in the form $x = g(x)$ suitable for fixed-point iteration.

since $x \neq 0$, $\frac{\cos(x)}{x} = 1 \iff g(x) = \cos(x) = x$

So $g(x) = \cos(x)$ would work

(b) Show that $g(x)$ maps the interval $[0, 1]$ into $[0, 1]$.

(c) Starting from $x_0 = 0.5$, perform three iterations of the fixed-point method $x_{n+1} = g(x_n)$. Write down each step.

(d) Discuss briefly whether the fixed-point iteration is expected to converge for this function and interval. Justify your answer by examining $|g'(x)|$ on $[0, 1]$.

(b) $\frac{d}{dx} g(x) = -\sin x \leqslant 0$ when $x \in [0,1]$ $\Rightarrow 0 = g(1) \leqslant g(x) \leqslant g(1) < g(\frac{\pi}{2}) = 1$ when $x \in [0,1]$

so $g(x)$ maps $[0,1]$ into $[0,1]$

(c) $x_1 = g(x_0) = \cos(0.5) = 0.87758$

$x_2 = g(x_1) = \cos(g(x_0)) = 0.639012$

$x_3 = g(x_2) = \cos(g(x_1)) = 0.802685$

(d) $g'(x) = -\sin x$

since $|g'(x)| < 1$ when $x \in [0,1]$

so for solution s and any initial point $x_0 \in [0,1]$

$|g'(s)| < 1$ and $|g'(x_0)| < 1$

$\therefore$ will converge

Problem 4 (20 points)

The function

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$$\frac{T_n}{T_{n-1}} = \frac{(2n^2-1)\cdot x^{2n}}{n!} \cdot \frac{(n-1)!}{[2(n-1)^2-1]\cdot x^{2n-2}} = \frac{2n^2-1}{[2(n-1)^2-1]n} \cdot x^2$$

$$\Rightarrow T_n = \begin{cases} \frac{2n^2-1}{[2(n-1)^2-1]n} \cdot x^2 \cdot T_{n-1} & (n \geqslant 2) \\ x^2 & (n=1) \end{cases}$$

by this way could avoid overflow.

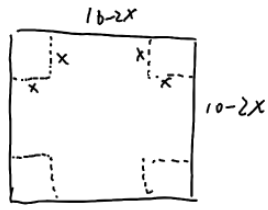
The code is attached in q4.m. Run function f will get the result of f(x)

by changing value of x on line 1, the code will output f(x)

P5.
the script is in q5.m, change the value of A and b in line 1,2, run and will output the solution x.

Problem 6 (20 points)

An open-top box is to be made by cutting squares of side length x from each corner of a 10×16 inch rectangular sheet of metal and folding up the sides. Determine the value(s) of x such that the volume of the box is exactly 100 cubic inches. Solve by implementing Newton's method in MATLAB.



the volume of the box is $(16-2x)\cdot(10-2x)\cdot x = 4x^3-52x^2+160x=100$

$\Rightarrow$ the problem is to solve

$$f(x) = 4x^3-52x^2+160x-100=0 \quad \text{where } x \in (0,5)$$

the Newton's method implementation is in q6.m.
By setting the x_0 in line 2 will set the initial value
Run the code will output the result s.

By setting $x_0=1$ will get $s_1 = 0.83902$
By setting $x_0=4$ will get $s_2 = 3.40175$